

2026 /03/ 15

1. Which of the following physical quantities is a scalar?
 - A. velocity
 - B. accelerated speed
 - C. displacement
 - D. distance travelled

2. The gravitational force acting on an object on the Moon is 30 N. Given that the Moon's gravitational acceleration is one-sixth of Earth's, what is the gravitational force acting on the same object on Earth?
 - A. 5 N
 - B. 30 N
 - C. 180 N
 - D. 300 N

3. A spring with an original length of 0.4 m and a stiffness coefficient $k = 15 \text{ N/m}$. When compressed by a force of 3 N, the new length of the spring is
 - A. 0.2 m
 - B. 0.4 m
 - C. 0.6 m
 - D. 0.8 m

4. When an object is subjected to two coplanar forces with magnitudes of 6 N and 8 N respectively, the resultant force may have a magnitude of
 - A. 0 N
 - B. 1 N
 - C. 10 N
 - D. 15 N

5. In a uniform magnetic field, a straight conductor with a length of 0.2 m is placed perpendicular to the magnetic field direction. When a current of 4 A flows through it, the resulting Ampere force is 0.8 N. Determine the magnetic induction intensity.
 - A. 0.1 T
 - B. 0.5 T
 - C. 1.0 T
 - D. 2.0 T

6. Which of the following statements about a point mass is correct?
 - A. When studying Earth's rotation, the Earth can be considered as a point mass.
 - B. When studying the time it takes for a train to pass through a tunnel, the train can be treated as a point mass.
 - C. When studying the high jump movement of athletes, the athlete can be considered as a point mass.
 - D. A point mass is an idealized model.

7. 摩天轮的半径为 R ，座舱随轮转动一周，其经过的路程和发生的位移分别为



A 路程 $2\pi R$ ，位移 $2R$

B 路程 $2R$ ，位移 $2\pi R$

C 路程 $2R$ ，位移 0

D 路程 $2\pi R$ ，位移 0

7. When a Ferris wheel with radius R rotates one full revolution, a passenger sitting on it travels through a **distance** and undergoes a **displacement** respectively of:

A Distance $2\pi R$, Displacement $2R$

B Distance $2R$, Displacement $2\pi R$

C Distance $2R$, Displacement 0

D Distance $2\pi R$, Displacement 0

8. Which of the following statements is correct?

A. The direction of sliding friction force is always opposite to the direction of object motion.

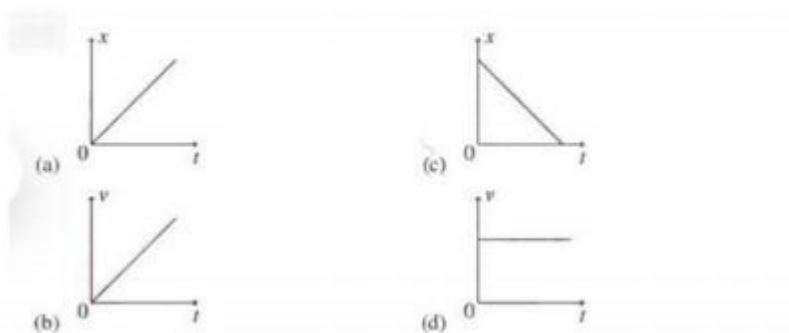
B. The magnitude of static friction force is directly proportional to the normal pressure.

C. The tighter the bottle is held, the greater the frictional force exerted on the bottle.

D. The frictional force exerted on the feet during walking is static friction.

9. Which of the following graphs can represent the state of a car moving with **uniformly accelerated linear motion** on a straight and level road?

9. 下列哪个图像能表示汽车在平直路面上做匀加速直线运动的状态



10. An object falls freely from a height of 80 m with $g = 10\text{ m/s}^2$. The time required to reach the ground is

A. 2 s

B. 4 s

C. 6 s

D. 8 s

11. An object undergoing uniform circular motion has a radius of 0.5 m and a period of 2 s . Its linear velocity is

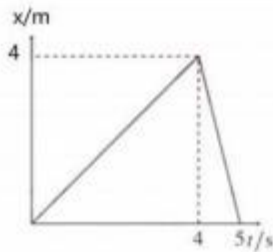
A. $0.5\pi\text{ m/s}$

B. $\pi\text{ m/s}$

C. $1.5\pi\text{ m/s}$

D. $2\pi\text{ m/s}$

12. 某玩具小车做直线运动的位移 - 时间 ($x-t$) 图像如图所示。下列说法正确的是



- A 0-4 秒内, 玩具小车的位移为 8 米 B 0-4 秒内, 玩具小车的速度为 1.0 米 / 秒
C 0-5 秒内, 玩具小车的平均速度为 0.8 米 / 秒 D 4-5 秒内, 玩具小车的速度为 0.8 米 / 秒

12. The position-time ($x-t$) graph of a toy car moving in a straight line is shown below. Which of the following statements is correct?

- A. In the time interval 0-4 s, the displacement of the toy car is 8 meters.
B. In the time interval 0-4 s, the speed of the toy car is 1.0 m/s
C. In the time interval 0-5 s, the average speed of the toy car is 0.8 m/s.
D. In the time interval 4-5 s, the speed of the toy car is 0.8 m/s.

13. The elevator decelerates downward with an acceleration of 1 m/s^2 , and the pressure exerted by a person with a mass of 50 kg on the elevator floor is ($g = 10 \text{ m/s}^2$).

- A. 450 N
B. 500 N
C. 550 N
D. 600 N

14. For an object with a mass of 5 kg sliding on a horizontal ground with a sliding friction force of 10 N, the dynamic friction coefficient is ($g = 10 \text{ m/s}^2$)

- A. 0.1
B. 0.2
C. 0.3
D. 0.4

15. On a smooth horizontal surface, an object with a mass of 3 kg is subjected to a constant force, and its velocity increases from 2 m/s to 8 m/s within 2 seconds. The magnitude of the force is

- A. 3 N
B. 6 N
C. 9 N
D. 12 N

16 光从空气射入某介质时, 入射角 $i=60^\circ$, 折射角 $r=30^\circ$ 。已知空气的折射率 $n=1$, 则该介质的折射率为

- A $\sqrt{3}$ B $\frac{\sqrt{3}}{3}$
C 1.5 D $\sqrt{5}$

16. When light travels from air into a certain medium, the angle of incidence is $i=60^\circ$ and the angle of refraction is $r=30^\circ$. Given that the refractive index of air is $n=1$, what is the refractive index of this medium?

- A. 3 B. 33 C. 1.5 D. 5

17. The small sphere undergoes uniform circular motion with an angular velocity of 200π rad/s and a radius of 0.05 m, resulting in a centripetal acceleration of magnitude

A. $2000\pi^2$ m/s²

B. $1000\pi^2$ m/s²

C. $200\pi^2$ m/s²

D. $2\pi^2$ /s

18. The spring oscillator performs simple harmonic motion with an amplitude of 5 cm and a period of 4s. Starting the timing from the equilibrium position, the distance traveled within 2s is

- A. 5 cm
- B. 10 cm
- C. 15 cm
- D. 20 cm

19. An object with a mass of 3 kg moving at a velocity of 4 m/s has a momentum magnitude of

- A. 7 kg·m/s
- B. 12 kg·m/s
- C. 24 kg·m/s
- D. 48 kg·m/s

20. A water drop falls freely from a height of 20 m without air resistance. The landing speed is ($g = 10 \text{ m/s}^2$).

- A. 10 m/s
- B. 15 m/s
- C. 20 m/s
- D. 25 m/s

21. When a horizontal object is pulled forward by a constant horizontal force F over a distance of 5 m, doing work of 50 J, the magnitude of F is

- A. 5 N
- B. 10 N
- C. 15 N
- D. 20 N

22. For a simple harmonic transverse wave with a wavelength of 8 m and a period of 2s, the wave velocity is

- A. 2 m/s
- B. 4 m/s
- C. 8 m/s
- D. 16 m/s

23. Two resistors with resistance values of 2Ω and 8Ω , respectively, have a total parallel resistance of

- A. 1.6Ω
- B. 2Ω
- C. 4Ω
- D. 10Ω

24. In the circuit diagram shown, $R = 4 \Omega$. When S is open, the voltage reading is 6 V; when S is closed, the reading is 4 V. Therefore, the internal resistance of the power supply is

- A. 0.5Ω
- B. 1Ω
- C. 2Ω
- D. 3Ω

25. For an ideal gas of a given mass, if the pressure remains constant and the volume increases, then the gas's

- A. temperature increment
- B. fall of temperature
- C. conservation of internal energy
- D. exothermic release

26. When an object is vertically thrown from the ground, the work done by gravity during its ascent is -100 J , then

- A. Kinetic energy decreased by 100 J
- B. The gravitational potential energy decreased by 100 J
- C. Mechanical energy decreased by 100 J
- D. Kinetic energy increased by 100 J

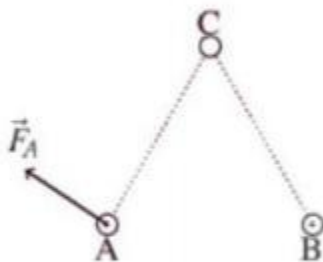
27. The current flowing through the conductor is 0.8 A , so the charge passing through the conductor's cross-section within 2 seconds is

- A. 0.4 C
- B. 1.6 C
- C. 2.4 C
- D. 3.2 C

28. For a fixed mass of ideal gas, if the temperature remains constant and the pressure decreases, what will happen to the volume of the gas?

- A. All of the above are possible
- B. Increase
- C. Decrease
- D. Remain unchanged

29. 三个点电荷 A、B、C 位于等边三角形的三个顶点，其中 A、B 均带正电。B、C 两个电荷对 A 的静电力的合力为 F_A ，如图所示。已知 F_A 与 BA 的延长线夹角小于 60° ，则关于电荷 C 的判断正确的是



- | | |
|--------------------|--------------------|
| A 带正电，电荷量小于 B 的电荷量 | B 带负电，电荷量大于 B 的电荷量 |
| C 带正电，电荷量大于 B 的电荷量 | D 带负电，电荷量小于 B 的电荷量 |

29. Three point charges A, B, and C are located at the vertices of an equilateral triangle. Charges A and B are both positively charged. The resultant electrostatic force exerted on A by charges B and C is F_A , as shown in the figure. It is known that the angle between F_A and the extension of line segment BA is less than 60° . Which of the following judgments about charge C is correct?

- A. Positively charged, with a magnitude of charge less than that of B
- B. Negatively charged, with a magnitude of charge greater than that of C. Positively charged, with a magnitude of charge greater than that of D. Negatively charged, with a magnitude of charge less than that of B

30. In a uniform electric field, which of the following statements is correct?

- A. The potentials at all points are equal
- B. The field strengths at all points are equal.
- C. The electric field lines are curves
- D. The direction of the electric field varies everywhere.

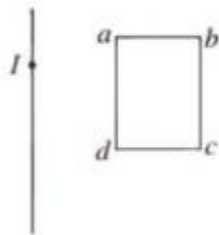
31. The wire is 0.5 m long, with a current of 3 A and a magnetic field of 0.2 T. The direction of the magnetic field forms a 60° angle with the current. The magnitude of the Ampere force is A. 0.15 N.

- B. 0.3 N
- C. 0.45 N
- D. 0.6 N

32. The coil area is 0.3 m^2 , the magnetic field is 0.5 T, the angle between the normal and the magnetic field direction is 30° , and the magnetic flux is

- A. 0.075 Wb
- B. 0.15 Wb
- C. 0.3 Wb
- D. 0.45 Wb

33 如图所示，一根通有电流 I 的长直导线右侧，固定一个矩形金属线框 $abcd$ ，线框的 cd 边与导线平行。调节电流 I ，使空间各点的磁感应强度随时间均匀增大，则



A 线框中的感应电流方向为 $a \rightarrow b \rightarrow c \rightarrow d \rightarrow a$

B 线框中的感应电流逐渐增大

C 线框受到的安培力方向水平向右

D 线框 cd 边受到的安培力大小保持不

33. As shown in the figure, a rectangular metal wire frame $abcd$ is fixed on the right side of a straight current-carrying wire carrying a current I . The side cd of the frame is parallel to the wire. Adjust the current I so that the magnetic induction intensity at all points in space increases uniformly with time. Then which of the following statements is correct?

- A. The direction of the induced current in the frame is $a \rightarrow b \rightarrow c \rightarrow d \rightarrow a$.
- B. The induced current in the frame increases gradually.
- C. The direction of the Ampère force experienced by the frame is horizontally to the right.
- D. The magnitude of the Ampère force acting on the side cd of the frame remains unchanged.

34 如图所示，空间存在电场， a 、 b 为电场中的两点。下列说法正确的是。



A b 点的电场强度大于 a 点的电场强度

B a 点的电场强度大于 b 点的电场强度

C 该电场为匀强电场

D a 、 b 两点的电场强度大小相等

34. As shown in the figure, there is an electric field in space, and points a and b are two points in this electric field. Which of the following statements is correct?

- A. The electric field strength at point b is greater than that at point
- B. The electric field strength at point a is greater than that at point

- C. This is a uniform electric field
- D. The electric field strength at points a and b is equal.

35. When a beam of light travels from air to glass at an incident angle of 45° and a refractive angle of 30° , the refractive index of the glass is

- A. $\sqrt{2}$
- B. $\sqrt{3}$
- C. 1.5
- D. 2

37. An object of mass m is thrown vertically upward from the ground with v_0 . Ignoring air resistance, the mechanical energy at the highest point is (with the ground as the zero potential energy surface).

- A. $\frac{1}{2}mv_0^2$
- B. mgh
- C. $\frac{1}{2}mv_0^2 + mgh$
- D. 0

38. The vehicle has a mass of 1.5 t and a power output of 45 kW. When traveling at a constant speed of 54 km/h, the resistance is

- A. 1000 N
- B. 1500 N
- C. 2000 N
- D. 3000 N

39. If the initial velocity of projectile motion increases to three times its original value, then

- A. The flight time has increased to three times the original duration.
- B. The horizontal displacement has tripled from the original value.
- C. landing speed remains constant
- D. Vertical displacement increases to three times the original value

40 如图所示，两个质量相等的物体 A、B 放在绕竖直轴匀速转动的水平圆盘上，且两物体相对圆盘保持静止，则



- A 物体 A 的角速度与物体 B 的角速度大小相等
- B 物体 A 的线速度与物体 B 的线速度大小相等
- C 物体 A 的角速度大于物体 B 的角速度
- D 物体 A 的角速度小于物体 B 的角速度

40. As shown in the figure, two objects A and B of equal mass are placed on a horizontal circular disk rotating uniformly about a vertical axis, and both objects remain stationary relative to the disk. Which of the following statements is correct?

- A. The angular velocity of object A is equal in magnitude to that of object B
- B. The linear velocity of object A is equal in magnitude to that of object B.
- C. The angular velocity of object A is greater than that of object B.
- E. The angular velocity of object A is less than that of object B.

41. The power of gravitational work done during the second second of free-fall motion ($g = 10 \text{ m/s}^2$, $m = 1 \text{ kg}$) is

- A. 50 W
- B. 100 W
- C. 150 W
- D. 200 W

42. The metal rod moves uniformly to the right along the guide rail, passing through the magnetic flux of the closed loop.
- A. magnify
 - B. abate
 - C. invariant
 - D. Increase first, then decrease

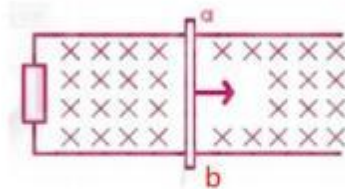
43. Two charges, $+q$ and $-2q$, are separated by a distance L . A charge q is placed on the extension line. After contact, the charges are released. The ratio of electrostatic forces is

- A. 2:1
- B. 1:2
- C. 9:4
- D. 4:9

44. A ball with a mass of 0.2 kg collides with a wall at 5 m/s , rebounds at 3 m/s , and has a contact time of 0.05 s . The average force is

- A. 8 N
- B. 16 N
- C. 24 N
- D. 32 N

46 如图所示，水平放置的平行金属导轨两端连接有电阻 R ，导线 ab 可在导轨上无摩擦滑动，匀强磁场垂直穿过导轨平面。当导线 ab 向右匀速滑动时，下列说法正确的是



- A 只要导线 ab 达到某一速度，撤去外力后，导线 ab 也能在导轨上保持匀速运动
- B 导线 ab 既受到拉力作用，又受到安培力作用
- C 导线 ab 的运动速度不变时，增大电阻 R 的阻值，拉动导线 ab 的拉力大小可以增大
- D 导线 ab 的运动速度越大，所需的拉力大小越小

46. As shown in the figure, a resistor R is connected to the ends of horizontally placed parallel metal guide rails. A conducting wire ab can slide on the rails without friction, and a uniform magnetic field passes perpendicularly through the plane of the rails. When the wire ab slides uniformly to the right, which of the following statements is correct?

- A. As long as the wire ab reaches a certain speed, after removing the external force, the wire ab can still maintain uniform motion on the rails.
- B. The wire ab is subjected to both a pulling force and an Ampère force.
- C. When the speed of the wire ab remains constant, increasing the resistance of R can the magnitude of the pulling force required to pull the wire ab .
- D. The greater the speed of the wire ab , the smaller the magnitude of the required pulling force.

48. The maximum acceleration of the vehicle is 4 m/s^2 , the speed is 72 km/h , the reaction time is 1 s , and the minimum safe distance is

- A. 20 m
- B. 30 m
- C. 40 m
- D. 50 m